

PROJECT TITLE: - BANKING FRAUD DETECTION & TRANSACTION ANALYTICS.

Prepared By: - Wani Umer.

Tools Used: -

- Microsoft Excel
- SQL
- Python (Pandas, NumPy, Matplotlib)
- Power BI
- DAX

EXECUTIVE SUMMARY

- This project was developed to analyze banking transaction data, identify fraud patterns, evaluate customer behavior, and provide actionable business insights through interactive dashboards.
- The solution integrates Excel, SQL, Python, and Power BI to transform raw transaction data into meaningful business intelligence. The report enables stakeholders to monitor fraud activity, analyze customer and transaction trends, and support data-driven decision-making.

BUSINESS OBJECTIVE

The primary objectives of this project are:

- Detect and analyze fraudulent transactions.
- Monitor fraud rates across transaction types.
- Understand customer transaction behavior.
- Identify trends in transaction volume and value.
- Build interactive dashboards for executive reporting.
- Provide recommendations to improve fraud monitoring and operational efficiency.

DATASET OVERVIEW

Attribute	Details
Domain	Banking & Financial Services
Dataset Type	Transactional Data
Records	50,000
Features	34
Target Variable	Fraud Status

Key Attributes

- Transaction ID
- Customer ID
- Transaction Amount
- Transaction Type
- Merchant Category
- Device Type
- Authentication Method
- Card Type
- Fraud Status
- Fraud Risk Score
- Transaction Date
- Time of Day
- Weekend Indicator
- Transaction Distance

PROJECT WORKFLOW

Raw Dataset.



Excel (Data Understanding).



SQL (Data Querying & Aggregation).



Python (Data Cleaning, Feature Engineering, EDA).



Power BI (Data Modeling, DAX, Dashboards).



Business Insights & Recommendations.

DATA PREPARATION

Excel

- Initial data exploration.
- Data validation.
- Basic cleaning.
- Understanding business attributes.

SQL

Performed analytical queries including:

- Fraud transaction analysis.
- Merchant category analysis.
- Customer segmentation.
- Transaction type analysis.
- Aggregation using GROUP BY.
- KPI calculations.

Python

Performed:

- Data cleaning.
- Missing value analysis.
- Duplicate detection.
- Feature engineering.
- Exploratory Data Analysis (EDA).
- Fraud investigation.
- Dashboard-ready dataset preparation.

DASHBOARD OVERVIEW

1. Executive Dashboard.

- Provides a high-level overview of banking operations and fraud metrics.

KPIs

- Total Transactions.
- Fraud Transactions.
- Total Transaction Value.
- Fraud Rate.
- Average Fraud Risk Score.
- Average Transaction Amount.

2. Fraud Analysis Dashboard

- Analyzes fraud patterns across multiple business dimensions.

Visuals

- Fraud by Transaction Type.
- Fraud by Merchant Category.
- Fraud by Authentication Method.
- Fraud by Device Type.
- Fraud by Time of Day.

3. Customer Behavior Dashboard

- Evaluates customer spending behavior and transaction trends.

KPIs

- Total Customers.
- Average Transaction Amount.
- Average Account Balance.
- Average Transactions per Customer.

4. Transaction Analysis Dashboard

- Analyzes transaction volume, value, and operational trends.

KPIs

- Total Transactions.
- Total Transaction Value.
- Average Transaction Amount.
- Average Daily Transactions.
- Average Transaction Distance.

5. Transaction Details Dashboard

- Provides detailed transaction records through Drill-through functionality for investigation and auditing.

6. Interactive Features

The Power BI report includes several interactive capabilities to enhance user experience and business analysis.

- Drill-through pages.
- Custom report tooltips.
- Page navigation buttons.
- Reset filter bookmarks.
- Cross-filtering between visuals.
- Interactive slicers.

BUSINESS INSIGHTS

1. Fraud transactions represent approximately **32%** of all transactions in the provided dataset.
2. Fraud activity is distributed across multiple transaction types, merchant categories, and authentication methods.
3. Customer transaction behavior remains relatively consistent across different card types.
4. Transaction values remain stable across quarterly periods due to the balanced synthetic dataset.
5. Interactive filtering enables rapid investigation of fraud patterns.

Note: The dataset used for this project is synthetic, resulting in intentionally balanced distributions across several categories.

BUSINESS RECOMMENDATIONS

1. Prioritize monitoring of high-risk transactions using the Fraud Risk Score.
2. Continuously monitor authentication methods for unusual fraud trends.
3. Implement real-time fraud alerts for high-risk transactions.
4. Review merchant categories showing higher fraud volumes.
5. Use interactive dashboards to support fraud investigation and executive decision-making.
6. Regularly review transaction trends to detect emerging fraud patterns.

PROJECT OUTCOMES

1. Business understanding.
2. Data preparation.
3. SQL analysis.
4. Python analysis.
5. Feature engineering.
6. Data visualization.
7. Dashboard development.
8. Business reporting.
9. Decision support.

SKILLS DEMONSTRATED

1. Analytics

- Data Cleaning
- Exploratory Data Analysis
- KPI Development
- Business Analysis
- Feature Engineering

2. Technical

- Excel
- SQL
- Python
- Power BI
- DAX

3. Business

- Fraud Analytics
- Executive Reporting
- Dashboard Design
- Data Storytelling
- Decision Support

FINAL ASSESSMENT

- This documentation is at the standard of a professional portfolio project and complements your dashboards by explaining the business context, analytical process, and outcomes. It gives recruiters and interviewers a clear narrative to follow rather than just a collection of visuals.